

Introduction to SQL and PostgreSQL

What is SQL?

SQL (Structured Query Language) is the standard language used to store, manage, and retrieve data in relational database systems.

It includes commands like:

- **SELECT** – retrieve data
- **INSERT** – add data
- **UPDATE** – modify data
- **DELETE** – remove data
- **WHERE, GROUP BY, ORDER BY, JOIN** – filter, group, sort, and connect tables

My notes cover all SQL fundamentals and advanced features.

What is PostgreSQL?

PostgreSQL is an **open-source, enterprise-grade database system** used worldwide.

Features:

- Supports all SQL operations
- Advanced data types (JSON, arrays, UUID)
- Triggers, functions, stored procedures
- Extensions (PostGIS, pg_stat_statements, etc.)
- MVCC for high performance

- Very stable and scalable

PostgreSQL Installation Steps

1. Install PostgreSQL on Windows

Step 1: Download

- Visit: <https://www.postgresql.org/download/windows>

Step 2: Install

- Run installer
- Select components:
 - ~ PostgreSQL Server
 - ~ pgAdmin 4
 - ~ Command Line Tools

Step 3: Set Superuser Password

- Create password for user **postgres**

Step 4: Keep Default Port

- Port: **5432**

Step 5: Finish Installation

Step 6: Open pgAdmin 4

- Enter your password
- Create a database → use Query Tool → run SQL queries

2. Install PostgreSQL on Linux

You can install PostgreSQL on any Linux distribution (Ubuntu, Debian, Fedora, CentOS, Arch).

Ubuntu / Debian (APT)

Step 1: Update system

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install PostgreSQL

```
sudo apt install postgresql postgresql-contrib -y
```

Step 3: Start & Enable Service

```
sudo systemctl start postgresql  
sudo systemctl enable postgresql
```

Step 4: Switch to PostgreSQL User

```
sudo -i -u postgres  
psql
```

Step 5: Create your own user

```
createuser --interactive
```

RedHat / CentOS / Fedora (YUM / DNF)

Step 1: Install

```
sudo dnf install postgresql-server postgresql-contrib
```

Step 2: Initialize Database

```
sudo postgresql-setup --initdb --unit postgresql
```

Step 3: Start Service

```
sudo systemctl start postgresql
```

```
sudo systemctl enable postgresql
```

3. Install PostgreSQL on macOS

Option 1: Install using Homebrew (Recommended)

Step 1: Update Brew

```
brew update
```

Step 2: Install PostgreSQL

```
brew install postgresql
```

Step 3: Start PostgreSQL

```
brew services start postgresql
```

Step 4: Login to PostgreSQL

```
psql postgres
```

Option 2: Install using Installer (Postgres.app)

Step 1: Download from:

<https://postgresapp.com/>

Step 2: Drag into Applications

Step 3: Open → PostgreSQL starts instantly

4. Install Using Docker (Windows / Linux / macOS)

```
docker run --name pgdb \  
-e POSTGRES_PASSWORD=12345 \  
-p 5432:5432 \  
-d postgres
```

This instantly sets up a PostgreSQL server in a container.

History of SQL

- Proposed by **E.F. Codd** (Relational Model – 1970).
- Developed by **IBM**: SEQUEL → SQL
- First commercial SQL version by Oracle (1979).
- Became ANSI standard in 1986, ISO standard in 1987.

How SQL Works (Process)

SQL query processing includes:

- Query Dispatcher
- Optimization Engine
- Classic Query Engine
- SQL Query Engine

These components find the fastest way to run your query.

Core SQL Commands (Very Important)

DDL – Data Definition Language

- **CREATE** – create table/database
- **DROP** – delete table/database

DML – Data Manipulation Language

- **INSERT** – add new records
- **UPDATE** – modify records
- **DELETE** – remove records

DQL – Data Query Language

- **SELECT** – fetch data

SQL vs NoSQL (Important for Exams / Interviews)

SQL

- Relational DB
- Fixed schema
- Tables (rows & columns)
- Uses ACID (reliable)
- Handles complex queries
- Examples: MySQL, PostgreSQL, Oracle, SQL Server

NoSQL

- Non-relational
- Dynamic schema (flexible)
- Documents, key-value, graphs
- BASE model
- Best for hierarchical & large-scale data
- Examples: MongoDB, Cassandra, Redis

Advantages of SQL

- Fast processing
- No coding needed
- Portable
- Standardized
- Multiple data views

Disadvantages

- Paid versions expensive
- Complex interface
- Partial control on some DB systems

SQL Syntax – Essentials

- Keywords **not case-sensitive**

- Usually written in **UPPERCASE**
- Based on relational algebra
- Simple English-like statements

Example:

```
SELECT column_name FROM table_name;
```

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PostgreSQL Data Types (Very Important)

A) Numeric Data Types

Type	Storage	Example
SMALLINT	2 bytes	-32768 to 32767
INT	4 bytes	-2147483648 to 2147483647
BIGINT	8 bytes	Very large numbers
REAL	4-byte float	10.45
FLOAT / DOUBLE	variable	234.345677
NUMERIC(p, s)	exact number	123.45

B) Monetary

Stores currency values

Range: **±92 quadrillion**

salary MONEY

C) Character Types

Type	Description
CHAR(n)	fixed length
VARCHAR(n)	variable length
TEXT	unlimited length

D) Binary

- **bytea** for storing images, files, encrypted data

E) Date/Time Types

Type	Stores
DATE	only date
TIME	only time
TIMESTAMP	date + time
TIMESTAMPTZ	date + time + timezone
INTERVAL	duration (years, days, time)

F) Boolean

Stores **TRUE** / **FALSE**

G) Enumerated (ENUM)

User-defined fixed values

Example:

```
CREATE TYPE months AS ENUM('JAN','FEB','MAR');
```

Download Complete SQL & PostgreSQL Notes + Practice Files

You can access the full set of **SQL/PostgreSQL notes** along with practice datasets and queries from this GitHub repository:

👉 [SQL-resources-and-tutorials by akshay-dhage](#)